

Warm-up

Question 1:

To find $P(1)$, we substitute $x = 1$ into the polynomial $P(x) = x^3 - 2x^2 + 5x + 1$.

$$\begin{aligned} P(1) &= (1)^3 - 2(1)^2 + 5(1) + 1 \\ &= 1 - 2 + 5 + 1 \\ &= 5 \end{aligned}$$

Question 2:

According to the Remainder Theorem, the remainder when a polynomial $P(x)$ is divided by $(x - c)$ is $P(c)$. Here, $P(x) = x^2 + 4x - 7$ and the divisor is $(x - 3)$, so $c = 3$.

$$\begin{aligned} \text{Remainder} &= P(3) \\ &= (3)^2 + 4(3) - 7 \\ &= 9 + 12 - 7 \\ &= 14 \end{aligned}$$

Question 3:

The Factor Theorem states that $(x - c)$ is a factor of $P(x)$ if and only if $P(c) = 0$. Given that $P(2) = 0$, we identify $c = 2$.

$$P(2) = 0 \implies (x - 2) \text{ is a factor of } P(x)$$

Comparing this to the options: (A) $x - 1$ (B) x (C) $x + 2$ (D) $x - 2$

The correct option is (D).

Question 4:

The degree of a polynomial is the highest power of x when the expression is fully expanded. Given $P(x) = (x^2 + 1)(x - 5)^3$.

$$\begin{aligned} \text{Degree of } (x^2 + 1) &= 2 \\ \text{Degree of } (x - 5)^3 &= 3 \\ \text{Total Degree} &= 2 + 3 \\ &= 5 \end{aligned}$$

Question 5:

Using the Remainder Theorem, the remainder when $P(x) = 2x^3 - x + 4$ is divided by $(x + 1)$ is $P(-1)$.

$$\begin{aligned} \text{Remainder} &= P(-1) \\ &= 2(-1)^3 - (-1) + 4 \\ &= 2(-1) + 1 + 4 \\ &= -2 + 5 \\ &= 3 \end{aligned}$$

Question 6:

To show that $(x - 4)$ is a factor of $P(x) = x^2 - 2x - 8$, we must show that $P(4) = 0$ by the Factor Theorem.

$$\begin{aligned} P(4) &= (4)^2 - 2(4) - 8 \\ &= 16 - 8 - 8 \\ &= 0 \end{aligned}$$

Since $P(4) = 0$, $(x - 4)$ is a factor of $P(x)$.

Question 7:

Substitute $x = 0$ into $P(x) = x^3 + kx - 10$.

$$\begin{aligned} P(0) &= (0)^3 + k(0) - 10 \\ &= -10 \end{aligned}$$

Question 8:

By the Factor Theorem, if $(x - c)$ is a factor of $Q(x)$, then $Q(c) = 0$. Here, the factor is $(x + 5)$, which can be written as $(x - (-5))$.

$$(x + 5) \text{ is a factor} \implies Q(-5) = 0$$

Question 9:

Given $f(x) = x^2 - 3$ and $g(x) = 2x + 1$, we find $h(x) = f(x) + g(x)$.

$$\begin{aligned} h(x) &= (x^2 - 3) + (2x + 1) \\ &= x^2 + 2x - 3 + 1 \\ &= x^2 + 2x - 2 \end{aligned}$$

Question 10:

Using the Remainder Theorem, the remainder when $P(x) = x^4 + 5$ is divided by $(x - 1)$ is $P(1)$.

$$\begin{aligned} \text{Remainder} &= P(1) \\ &= (1)^4 + 5 \\ &= 1 + 5 \\ &= 6 \end{aligned}$$

Skill-building

Question 1:

Given $P(x) = x^3 + kx^2 - 4$ and the remainder when divided by $(x - 2)$ is 8. By the Remainder Theorem, $P(2) = 8$.

$$\begin{aligned}
P(2) &= (2)^3 + k(2)^2 - 4 \\
8 &= 8 + 4k - 4 \\
8 &= 4k + 4 \\
4 &= 4k \\
k &= 1
\end{aligned}$$

Question 2:

First, we show $(x + 2)$ is a factor of $P(x) = x^3 + 3x^2 - 4$ by checking if $P(-2) = 0$.

$$\begin{aligned}
P(-2) &= (-2)^3 + 3(-2)^2 - 4 \\
&= -8 + 3(4) - 4 \\
&= -8 + 12 - 4 \\
&= 0
\end{aligned}$$

Since $P(-2) = 0$, $(x + 2)$ is a factor. Now we divide $P(x)$ by $(x + 2)$ using polynomial division or synthetic division.

$$\begin{aligned}
&\text{Dividing } x^3 + 3x^2 - 4 \text{ by } (x + 2) : \\
&\text{Quotient} = x^2 + x - 2
\end{aligned}$$

Thus, $P(x) = (x + 2)(x^2 + x - 2)$.

Question 3:

If $P(x) = x^3 + mx + n$ is divisible by $(x - 1)$ and $(x + 1)$, then $P(1) = 0$ and $P(-1) = 0$.

$$\begin{aligned}
P(1) = 0 &\implies (1)^3 + m(1) + n = 0 \\
&\implies 1 + m + n = 0 \\
&\implies m + n = -1 \quad (1)
\end{aligned}$$

$$\begin{aligned}
P(-1) = 0 &\implies (-1)^3 + m(-1) + n = 0 \\
&\implies -1 - m + n = 0 \\
&\implies -m + n = 1 \quad (2)
\end{aligned}$$

Adding equations (1) and (2):

$$\begin{aligned}
(m + n) + (-m + n) &= -1 + 1 \\
2n &= 0 \\
n &= 0
\end{aligned}$$

Substituting $n = 0$ into (1):

$$\begin{aligned}
m + 0 &= -1 \\
m &= -1
\end{aligned}$$

Question 4:

To determine if $(2x - 1)$ is a factor of $P(x) = 4x^3 - 4x^2 + x$, we check if $P(1/2) = 0$.

$$\begin{aligned}
P(1/2) &= 4(1/2)^3 - 4(1/2)^2 + (1/2) \\
&= 4(1/8) - 4(1/4) + 1/2 \\
&= 1/2 - 1 + 1/2 \\
&= 0
\end{aligned}$$

Since $P(1/2) = 0$, $(2x - 1)$ is a factor of $P(x)$.

Question 5:

Given $P(x) = x^3 - 7x + 6$. The constant term is 6. Possible integer roots are factors of 6: $\pm 1, \pm 2, \pm 3, \pm 6$.

$$P(1) = 1^3 - 7(1) + 6 = 1 - 7 + 6 = 0 \implies x = 1 \text{ is a root.}$$

$$P(2) = 2^3 - 7(2) + 6 = 8 - 14 + 6 = 0 \implies x = 2 \text{ is a root.}$$

$$P(-3) = (-3)^3 - 7(-3) + 6 = -27 + 21 + 6 = 0 \implies x = -3 \text{ is a root.}$$

Integer roots are $x = 1, 2, -3$.

Question 6:

Divide $x^3 - 2x^2 + x - 5$ by $(x - 3)$ using long division.

$$\begin{aligned}
&\text{Step 1: } x^3/x = x^2 \\
&\quad (x^2)(x - 3) = x^3 - 3x^2 \\
&(x^3 - 2x^2 + x - 5) - (x^3 - 3x^2) = x^2 + x - 5 \\
&\quad \text{Step 2: } x^2/x = x \\
&\quad \quad (x)(x - 3) = x^2 - 3x \\
&(x^2 + x - 5) - (x^2 - 3x) = 4x - 5 \\
&\quad \text{Step 3: } 4x/x = 4 \\
&\quad \quad (4)(x - 3) = 4x - 12 \\
&(4x - 5) - (4x - 12) = 7
\end{aligned}$$

Quotient: $x^2 + x + 4$, Remainder: 7.

Question 7:

Given $f(x) = ax^3 + 3x^2 + b$. Since $(x - 1)$ is a factor, $f(1) = 0$.

$$\begin{aligned}
f(1) &= a(1)^3 + 3(1)^2 + b = 0 \\
\implies a + 3 + b &= 0 \\
\implies a + b &= -3 \quad (1)
\end{aligned}$$

Given $f(2) = 27$.

$$\begin{aligned}
f(2) &= a(2)^3 + 3(2)^2 + b = 27 \\
\implies 8a + 12 + b &= 27 \\
\implies 8a + b &= 15 \quad (2)
\end{aligned}$$

Subtract (1) from (2):

$$\begin{aligned}(8a + b) - (a + b) &= 15 - (-3) \\ 7a &= 18 \\ a &= 18/7\end{aligned}$$

Substitute $a = 18/7$ into (1):

$$\begin{aligned}18/7 + b &= -3 \\ b &= -3 - 18/7 = -21/7 - 18/7 = -39/7\end{aligned}$$

Question 8:

The division algorithm for polynomials states that $P(x) = D(x)Q(x) + R(x)$, where $D(x)$ is the divisor and $Q(x)$ is the quotient. Given $P(x) = (x - k)Q(x) + R$.

Substitute $x = k$:

$$\begin{aligned}P(k) &= (k - k)Q(k) + R \\ &= 0 \cdot Q(k) + R \\ &= R\end{aligned}$$

Question 9:

Given $P(x) = x^3 - 19x - 30$ and $P(-2) = 0$. By the Factor Theorem, $(x + 2)$ is a factor.

$$\begin{aligned}\text{Divide } P(x) \text{ by } (x + 2) : \\ \text{Quotient} &= x^2 - 2x - 15\end{aligned}$$

Now factorise the quadratic $x^2 - 2x - 15$.

$$\begin{aligned}\text{Looking for numbers that multiply to } -15 \text{ and add to } -2 : \\ -5 \text{ and } 3\end{aligned}$$

$$\text{Quadratic factor} = (x - 5)(x + 3)$$

Thus, $P(x) = (x + 2)(x - 5)(x + 3)$.

Question 10:

Let the remainder be $R(x)$. Since the divisor is $(x - 1)(x + 2)$, which is a quadratic, the remainder must be linear: $R(x) = ax + b$.

$$P(x) = (x - 1)(x + 2)Q(x) + (ax + b)$$

Using the given remainders:

$$\begin{aligned}P(1) = 5 &\implies a(1) + b = 5 \\ \implies a + b &= 5 \quad (1)\end{aligned}$$

$$\begin{aligned} P(-2) &= -1 \implies a(-2) + b = -1 \\ \implies -2a + b &= -1 \quad (2) \end{aligned}$$

Subtract (2) from (1):

$$\begin{aligned} (a + b) - (-2a + b) &= 5 - (-1) \\ 3a &= 6 \\ a &= 2 \end{aligned}$$

Substitute $a = 2$ into (1):

$$\begin{aligned} 2 + b &= 5 \\ b &= 3 \end{aligned}$$

Therefore, the remainder is $R(x) = 2x + 3$.

Easier Exam Questions

Question 1:

Given $P(x) = -x^3 - 2x^2 - 3x + 8$ and the divisor is $x + 2$. By the Remainder Theorem, the remainder is $P(-2)$.

$$\begin{aligned} P(-2) &= -(-2)^3 - 2(-2)^2 - 3(-2) + 8 \\ &= -(-8) - 2(4) + 6 + 8 \\ &= 8 - 8 + 6 + 8 \\ &= 14 \end{aligned}$$

The correct option is (D).

Question 2:

Part (i): Given $P(x) = 2x^3 - x^2 - 16x + a$ and $(x + 3)$ is a factor, so $P(-3) = 0$.

$$\begin{aligned} P(-3) &= 2(-3)^3 - (-3)^2 - 16(-3) + a = 0 \\ &= 2(-27) - 9 + 48 + a = 0 \\ &= -54 - 9 + 48 + a = 0 \\ &= -15 + a = 0 \\ a &= 15 \end{aligned}$$

Part (ii): Now $P(x) = 2x^3 - x^2 - 16x + 15$. We know $(x + 3)$ is a factor. Divide $P(x)$ by $(x + 3)$.

$$\begin{aligned} \text{Dividing } 2x^3 - x^2 - 16x + 15 \text{ by } (x + 3) : \\ \text{Quotient} &= 2x^2 - 7x + 5 \end{aligned}$$

Now factorise the quadratic $2x^2 - 7x + 5$.

Looking for numbers that multiply to 10 and add to -7 :
 -2 and -5

$$\text{Quadratic factor} = (2x - 5)(x - 1)$$

Thus, $P(x) = (x + 3)(2x - 5)(x - 1)$.

Question 3:

Part (i): Given $P(x) = 2x^3 + 3x^2 - 29x - 60$. To show $(x + 3)$ is a factor, check $P(-3)$.

$$\begin{aligned} P(-3) &= 2(-3)^3 + 3(-3)^2 - 29(-3) - 60 \\ &= 2(-27) + 3(9) + 87 - 60 \\ &= -54 + 27 + 87 - 60 \\ &= 0 \end{aligned}$$

Since $P(-3) = 0$, $(x + 3)$ is a factor.

Part (ii): Divide $P(x)$ by $(x + 3)$.

$$\begin{aligned} \text{Dividing } 2x^3 + 3x^2 - 29x - 60 \text{ by } (x + 3) : \\ \text{Quotient} = 2x^2 - 3x - 20 \end{aligned}$$

Now factorise $2x^2 - 3x - 20$.

Looking for numbers that multiply to -40 and add to -3 :
 -8 and 5

$$\text{Quadratic factor} = (2x + 5)(x - 4)$$

Thus, $P(x) = (x + 3)(2x + 5)(x - 4)$.

Question 4:

Part (i): Given $P(x) = 3x^3 + 2x^2 - 7x + 2$. To show $(x - 1)$ is a factor, check $P(1)$.

$$\begin{aligned} P(1) &= 3(1)^3 + 2(1)^2 - 7(1) + 2 \\ &= 3 + 2 - 7 + 2 \\ &= 0 \end{aligned}$$

Since $P(1) = 0$, $(x - 1)$ is a factor.

Part (ii): Divide $P(x)$ by $(x - 1)$.

$$\begin{aligned} \text{Dividing } 3x^3 + 2x^2 - 7x + 2 \text{ by } (x - 1) : \\ \text{Quotient} = 3x^2 + 5x - 2 \end{aligned}$$

Now factorise $3x^2 + 5x - 2$.

Looking for numbers that multiply to -6 and add to 5 :
 6 and -1

$$\text{Quadratic factor} = (3x - 1)(x + 2)$$

Thus, $P(x) = (x - 1)(3x - 1)(x + 2)$.

Question 5:

Given $P(x) = x^3 - 2x^2 + x + 5$ and the divisor is $(x - 2)$. By the Remainder Theorem, the remainder is $P(2)$.

$$\begin{aligned} P(2) &= (2)^3 - 2(2)^2 + (2) + 5 \\ &= 8 - 8 + 2 + 5 \\ &= 7 \end{aligned}$$

Question 6:

Given $f(x) = x^2 + 5$ and $g(x) = -4x$.

Part (a): Find $g(x) - f(x)$.

$$\begin{aligned} g(x) - f(x) &= (-4x) - (x^2 + 5) \\ &= -x^2 - 4x - 5 \end{aligned}$$

Part (b): Degree of $f(x) \times g(x)$.

$$\begin{aligned} \text{Degree of } f(x) &= 2 \\ \text{Degree of } g(x) &= 1 \\ \text{Total Degree} &= 2 + 1 = 3 \end{aligned}$$

Part (c): Find $f(g(x))$.

$$\begin{aligned} f(g(x)) &= f(-4x) \\ &= (-4x)^2 + 5 \\ &= 16x^2 + 5 \end{aligned}$$

Question 7:

Part (i): Given $P(x) = x^3 - 6x^2 + 10x - 3$. Show $P(3) = 0$.

$$\begin{aligned} P(3) &= (3)^3 - 6(3)^2 + 10(3) - 3 \\ &= 27 - 54 + 30 - 3 \\ &= 0 \end{aligned}$$

Part (ii): Since $P(3) = 0$, $(x - 3)$ is a factor. Divide $P(x)$ by $(x - 3)$.

$$\begin{aligned} \text{Dividing } x^3 - 6x^2 + 10x - 3 \text{ by } (x - 3) : \\ \text{Quotient} &= x^2 - 3x + 1 \end{aligned}$$

Now solve $x^2 - 3x + 1 = 0$ using the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

$$\begin{aligned}
x &= \frac{-(-3) \pm \sqrt{(-3)^2 - 4(1)(1)}}{2(1)} \\
&= \frac{3 \pm \sqrt{9 - 4}}{2} \\
&= \frac{3 \pm \sqrt{5}}{2}
\end{aligned}$$

Thus, the solutions to $P(x) = 0$ are $x = 3, \frac{3+\sqrt{5}}{2}, \frac{3-\sqrt{5}}{2}$.

Harder Exam Questions

Question 1:

Part (i): Given $P(x) = x^3 + 2x^2 - 5x + 1$. To show the remainder of $P(x) \div (x - 2)$ is 7, we find $P(2)$.

$$\begin{aligned}
P(2) &= (2)^3 + 2(2)^2 - 5(2) + 1 \\
&= 8 + 8 - 10 + 1 \\
&= 7
\end{aligned}$$

Part (ii): Verify using polynomial division.

Dividing $x^3 + 2x^2 - 5x + 1$ by $(x - 2)$:

Step 1: $x^3/x = x^2$

$$(x^2)(x - 2) = x^3 - 2x^2$$

$$(x^3 + 2x^2 - 5x + 1) - (x^3 - 2x^2) = 4x^2 - 5x + 1$$

Step 2: $4x^2/x = 4x$

$$(4x)(x - 2) = 4x^2 - 8x$$

$$(4x^2 - 5x + 1) - (4x^2 - 8x) = 3x + 1$$

Step 3: $3x/x = 3$

$$(3)(x - 2) = 3x - 6$$

$$(3x + 1) - (3x - 6) = 7$$

Thus, the remainder is 7.

Question 2:

Part (i): For $2x^4 + mx^2 - 2$ to be divisible by $x + 2$, the remainder $P(-2)$ must be 0.

$$\begin{aligned}
P(-2) &= 2(-2)^4 + m(-2)^2 - 2 = 0 \\
&= 2(16) + 4m - 2 = 0 \\
&= 32 + 4m - 2 = 0 \\
&= 30 + 4m = 0 \\
4m &= -30 \\
m &= -7.5
\end{aligned}$$

Part (ii): Divide $P(x) = x^4 - x^3 + x^2 - x + 1$ by $x^2 + 4$.

$$\begin{aligned}
&\text{Step 1: } x^4/x^2 = x^2 \\
&\quad (x^2)(x^2 + 4) = x^4 + 4x^2 \\
&(x^4 - x^3 + x^2 - x + 1) - (x^4 + 4x^2) = -x^3 - 3x^2 - x + 1 \\
&\text{Step 2: } -x^3/x^2 = -x \\
&\quad (-x)(x^2 + 4) = -x^3 - 4x \\
&(-x^3 - 3x^2 - x + 1) - (-x^3 - 4x) = -3x^2 + 3x + 1 \\
&\text{Step 3: } -3x^2/x^2 = -3 \\
&\quad (-3)(x^2 + 4) = -3x^2 - 12 \\
&(-3x^2 + 3x + 1) - (-3x^2 - 12) = 3x + 13
\end{aligned}$$

Quotient: $x^2 - x - 3$, Remainder: $3x + 13$.

Part (iii): For $x^4 - x^3 + x^2 + ax + b$ to be divisible by $x^2 + 4$, the remainder must be 0.

Dividing $x^4 - x^3 + x^2 + ax + b$ by $x^2 + 4$:

$$\begin{aligned}
&\text{Step 1: } x^4/x^2 = x^2 \\
&\quad (x^2)(x^2 + 4) = x^4 + 4x^2 \\
&(x^4 - x^3 + x^2 + ax + b) - (x^4 + 4x^2) = -x^3 - 3x^2 + ax + b \\
&\text{Step 2: } -x^3/x^2 = -x \\
&\quad (-x)(x^2 + 4) = -x^3 - 4x \\
&(-x^3 - 3x^2 + ax + b) - (-x^3 - 4x) = -3x^2 + (a + 4)x + b \\
&\text{Step 3: } -3x^2/x^2 = -3 \\
&\quad (-3)(x^2 + 4) = -3x^2 - 12 \\
&(-3x^2 + (a + 4)x + b) - (-3x^2 - 12) = (a + 4)x + (b + 12)
\end{aligned}$$

For the remainder to be 0, we must have:

$$\begin{aligned}
a + 4 = 0 &\implies a = -4 \\
b + 12 = 0 &\implies b = -12
\end{aligned}$$

Question 3:

Part (i): Given $f(x) = x(x + 1) - a(a + 1)$. We look for a factor using the Factor Theorem. Let's test $x = a$.

$$f(a) = a(a + 1) - a(a + 1) = 0$$

Since $f(a) = 0$, $(x - a)$ is a factor.

Part (ii): Now we divide $f(x)$ by $(x - a)$.

$$\begin{aligned}
f(x) &= x^2 + x - (a^2 + a) \\
&= x^2 + x - a^2 - a \\
&= (x^2 - a^2) + (x - a) \\
&= (x - a)(x + a) + 1(x - a) \\
&= (x - a)(x + a + 1)
\end{aligned}$$

Thus, $f(x) = (x - a)(x + a + 1)$.

Question 4:

Given $P(x) = x^3 - 3x^2 + kx - 5$. When divided by $(x + 2)$, the remainder is 7. By the Remainder Theorem, $P(-2) = 7$.

$$\begin{aligned}
 P(-2) &= (-2)^3 - 3(-2)^2 + k(-2) - 5 = 7 \\
 &= -8 - 3(4) - 2k - 5 = 7 \\
 &= -8 - 12 - 2k - 5 = 7 \\
 &= -25 - 2k = 7 \\
 &= -2k = 32 \\
 k &= -16
 \end{aligned}$$

Question 5:

Given $P(x) = x^4 - 2x^3 - kx + 27$ and $(x - 2)$ is a factor, so $P(2) = 0$.

$$\begin{aligned}
 P(2) &= (2)^4 - 2(2)^3 - k(2) + 27 = 0 \\
 &= 16 - 16 - 2k + 27 = 0 \\
 &= -2k + 27 = 0 \\
 2k &= 27 \\
 k &= 13.5
 \end{aligned}$$

Question 6:

Part (i): Find linear factors of $P(x) = x^3 - 5x^2 + 8x - 4$. Test factors of -4 : $\pm 1, \pm 2, \pm 4$.

$$P(1) = 1 - 5 + 8 - 4 = 0 \implies (x - 1) \text{ is a factor.}$$

Divide $P(x)$ by $(x - 1)$:

$$\text{Quotient} = x^2 - 4x + 4 = (x - 2)^2$$

Thus, the linear factors are $(x - 1)(x - 2)(x - 2)$.

Part (ii): Solve $x^3 - 5x^2 + 8x - 4 > 0$.

$$(x - 1)(x - 2)^2 > 0$$

Since $(x - 2)^2 \geq 0$ for all x , the inequality depends on $(x - 1)$. However, $(x - 2)^2$ cannot be 0 because the inequality is strict.

$$\begin{aligned}
 x - 1 &> 0 \text{ and } x \neq 2 \\
 x &> 1 \text{ and } x \neq 2
 \end{aligned}$$

Solution: $x \in (1, 2) \cup (2, \infty)$.

Question 7:

Given $P(x) = 2x^3 + ax + 4$ and a zero at $x = -2$, so $P(-2) = 0$.

$$\begin{aligned}
P(-2) &= 2(-2)^3 + a(-2) + 4 = 0 \\
&= 2(-8) - 2a + 4 = 0 \\
&= -16 - 2a + 4 = 0 \\
&= -12 - 2a = 0 \\
2a &= -12 \\
a &= -6
\end{aligned}$$

Question 8:

Part (i): Sketch $y = x(x+3)(x-2)^2$.

- x-intercepts: $x = 0, -3, 2$.
- At $x = 2$, the graph touches the x-axis (multiplicity 2).
- As $x \rightarrow \infty, y \rightarrow \infty$; as $x \rightarrow -\infty, y \rightarrow \infty$ (degree 4, positive leading coefficient).

Part (ii): Solve $x(x+3)(x-2)^2 \geq 0$.

Critical points: $x = -3, 0, 2$.

Interval $(-\infty, -3] : y \geq 0$ (e.g., $x = -4 \implies (-4)(-1)(-6)^2 > 0$)

Interval $[-3, 0] : y \leq 0$ (e.g., $x = -1 \implies (-1)(2)(-3)^2 < 0$)

Interval $[0, 2] : y \geq 0$ (e.g., $x = 1 \implies (1)(4)(-1)^2 > 0$)

Interval $[2, \infty) : y \geq 0$ (e.g., $x = 3 \implies (3)(6)(1)^2 > 0$)

Solution: $x \in (-\infty, -3] \cup [0, \infty)$.

Question 9:

Given $P(x) = x^2 + ax + b$. Zero at $x = 2 \implies P(2) = 0$.

$$\begin{aligned}
P(2) &= (2)^2 + a(2) + b = 0 \\
\implies 4 + 2a + b &= 0 \\
\implies 2a + b &= -4 \quad (1)
\end{aligned}$$

Remainder when divided by $x + 1$ is 18 $\implies P(-1) = 18$.

$$\begin{aligned}
P(-1) &= (-1)^2 + a(-1) + b = 18 \\
\implies 1 - a + b &= 18 \\
\implies -a + b &= 17 \quad (2)
\end{aligned}$$

Subtract (2) from (1):

$$\begin{aligned}(2a + b) - (-a + b) &= -4 - 17 \\ 3a &= -21 \\ a &= -7\end{aligned}$$

Substitute $a = -7$ into (2):

$$\begin{aligned}\implies -(-7) + b &= 17 \\ \implies 7 + b &= 17 \\ b &= 10\end{aligned}$$

Question 10:

Given $f(x) = px^3 - 4x^2 - 11x + q$. Factors $(x - 2)$ and $(2x + 3)$.

$$\begin{aligned}f(2) = 0 &\implies p(2)^3 - 4(2)^2 - 11(2) + q = 0 \\ \implies 8p - 16 - 22 + q &= 0 \\ \implies 8p + q &= 38 \quad (1)\end{aligned}$$

$$\begin{aligned}f(-3/2) = 0 &\implies p(-3/2)^3 - 4(-3/2)^2 - 11(-3/2) + q = 0 \\ \implies p(-27/8) - 4(9/4) + 33/2 + q &= 0 \\ \implies -27p/8 - 9 + 16.5 + q &= 0 \\ \implies -27p/8 + q &= -7.5 \quad (2)\end{aligned}$$

Subtract (2) from (1):

$$\begin{aligned}(8p + q) - (-27p/8 + q) &= 38 - (-7.5) \\ (64p/8 + 27p/8) &= 45.5 \\ 91p/8 &= 45.5 \\ 91p &= 364 \\ p &= 4\end{aligned}$$

Substitute $p = 4$ into (1):

$$\begin{aligned}8(4) + q &= 38 \\ 32 + q &= 38 \\ q &= 6\end{aligned}$$

Now $f(x) = 4x^3 - 4x^2 - 11x + 6$. Divide by $(x - 2)(2x + 3) = 2x^2 + 3x - 4x - 6 = 2x^2 - x - 6$.

$$\begin{aligned}\text{Dividing } 4x^3 - 4x^2 - 11x + 6 \text{ by } 2x^2 - x - 6 : \\ \text{Quotient} &= 2x - 1\end{aligned}$$

Thus, $f(x) = (x - 2)(2x + 3)(2x - 1)$.

Question 11:

Given $P(x) = x^2 - 2ax + a^4$ and $Q(x) = bx^2 + x + 1$.

Condition 1: $P(x)$ divided by $x + b$ leaves remainder 1.

$$\begin{aligned} P(-b) = 1 &\implies (-b)^2 - 2a(-b) + a^4 = 1 \\ &\implies b^2 + 2ab + a^4 = 1 \quad (1) \end{aligned}$$

Condition 2: $Q(x)$ has a factor $(ax - 1)$. This means $Q(1/a) = 0$.

$$\begin{aligned} Q(1/a) = 0 &\implies b(1/a)^2 + (1/a) + 1 = 0 \\ &\implies b/a^2 + 1/a + 1 = 0 \\ &\implies b + a + a^2 = 0 \\ &\implies b = -a^2 - a \quad (2) \end{aligned}$$

Substitute (2) into (1):

$$\begin{aligned} &(-a^2 - a)^2 + 2a(-a^2 - a) + a^4 = 1 \\ \implies &(a^4 + 2a^3 + a^2) - 2a^3 - 2a^2 + a^4 = 1 \\ &\implies 2a^4 - a^2 - 1 = 0 \end{aligned}$$

Let $u = a^2$. Then $2u^2 - u - 1 = 0$.

$$(2u + 1)(u - 1) = 0 \implies u = -1/2 \text{ or } u = 1$$

Since a is a real number, $a^2 = 1 \implies a = \pm 1$.

Case 1: $a = 1$.

$$b = -(1)^2 - 1 = -2$$

Case 2: $a = -1$.

$$b = -(-1)^2 - (-1) = -1 + 1 = 0$$

Since a is non-zero and b is typically implied to be non-zero for $Q(x)$ to be quadratic, we check $b = 0$. If $b = 0$, $Q(x) = x + 1$, which has factor $(ax - 1)$ if $a = -1$. However, the question asks for b . If $b = 0$, $Q(x)$ is linear. Usually, $b \neq 0$. If $b = 0$, $a = -1$. If $b = -2$, $a = 1$.

Checking $b = 0$ in $P(x)$ remainder: $P(0) = a^4 = 1 \implies a = \pm 1$. This is consistent.

Values of b are 0, -2.

Question 12:

To show $(x - 1)(x - 2)$ is a factor of $P(x)$, we must show $P(1) = 0$ and $P(2) = 0$.

Check $P(1)$:

$$\begin{aligned} P(1) &= 1^n(2^m - 1) + 1^m(1 - 2^n) + (2^n - 2^m) \\ &= 2^m - 1 + 1 - 2^n + 2^n - 2^m \\ &= 0 \end{aligned}$$

Check $P(2)$:

$$\begin{aligned}
P(2) &= 2^n(2^m - 1) + 2^m(1 - 2^n) + (2^n - 2^m) \\
&= 2^{n+m} - 2^n + 2^m - 2^{m+n} + 2^n - 2^m \\
&= 0
\end{aligned}$$

Since $P(1) = 0$ and $P(2) = 0$, $(x - 1)$ and $(x - 2)$ are factors. Thus, $(x - 1)(x - 2)$ is a factor of $P(x)$.

Question 13:

Given $P(x) = x^3 + ax^2 + bx + 6$.

Condition 1: $(x - 2)$ is a factor $\implies P(2) = 0$.

$$\begin{aligned}
P(2) &= (2)^3 + a(2)^2 + b(2) + 6 = 0 \\
\implies 8 + 4a + 2b + 6 &= 0 \\
\implies 4a + 2b &= -14 \\
\implies 2a + b &= -7 \quad (1)
\end{aligned}$$

Condition 2: Remainder when divided by $(x - 1)$ is 10 $\implies P(1) = 10$.

$$\begin{aligned}
P(1) &= (1)^3 + a(1)^2 + b(1) + 6 = 10 \\
\implies 1 + a + b + 6 &= 10 \\
\implies a + b &= 3 \quad (2)
\end{aligned}$$

Subtract (2) from (1):

$$\begin{aligned}
(2a + b) - (a + b) &= -7 - 3 \\
a &= -10
\end{aligned}$$

Substitute $a = -10$ into (2):

$$\begin{aligned}
-10 + b &= 3 \\
b &= 13
\end{aligned}$$

Question 14:

Given $P(x) = x^3 + x^2 + ax + b$.

Condition 1: $(x - 1)$ is a factor $\implies P(1) = 0$.

$$\begin{aligned}
P(1) &= (1)^3 + (1)^2 + a(1) + b = 0 \\
\implies 1 + 1 + a + b &= 0 \\
\implies a + b &= -2 \quad (1)
\end{aligned}$$

Condition 2: Remainder when divided by $(x - 2)$ is 10 $\implies P(2) = 10$.

$$\begin{aligned}
P(2) &= (2)^3 + (2)^2 + a(2) + b = 10 \\
\implies 8 + 4 + 2a + b &= 10 \\
\implies 12 + 2a + b &= 10 \\
\implies 2a + b &= -2 \quad (2)
\end{aligned}$$

Subtract (1) from (2):

$$\begin{aligned}(2a + b) - (a + b) &= -2 - (-2) \\ a &= 0\end{aligned}$$

Substitute $a = 0$ into (1):

$$\begin{aligned}0 + b &= -2 \\ b &= -2\end{aligned}$$